

HİLAL İLHAN

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Education

Boğaziçi University Sep 2021 – May 2027
BSc in Electrical and Electronics Engineering İstanbul,Turkey

- GPA: 3.65/4.00
- Specialization in **Control Systems** and **Signal Processing**
- Minor in Economics

Technical University of Munich Mar – Sep 2025
Erasmus+ Exchange Program Munich, Germany

Anadolu University Sep 2021 – Jun 2023
Photography and Camerawork (Open Education Faculty)

Experience

Mercedes-Benz – Central Electronics & Classical HMI Oct 2026 – Present
Research & Development – PEP Intern İstanbul,Turkey

- Supporting global R&D activities involving Door Control Modules and Anti-Theft Alarm Systems, including systems analysis, requirements evaluation, and cross-functional collaboration.

Koç University – Robotics and Mechatronics Laboratory Jan 2026 – Present
Undergraduate Researcher İstanbul,Turkey

- Conducting research under the supervision of **Prof. Çağatay Başdoğan**, in collaboration with Boğaziçi University's Automatic Control and Microprocessors Laboratory.

National Taiwan University – Next-generation Automated Surgical Apparatus Lab Jul – Sep 2026
Research Intern Taipei, Taiwan

- Conducted research on **learning-based** and **data-driven control** under the supervision of **Prof. Cheng-Wei Chen**.

Technical University of Berlin – Distributed AI Lab Jul – Sep 2024
Robotics Engineer Intern Berlin, Germany

- Worked on **SLAM** and **ROS 2** under the supervision of **Prof. Şahin Albayrak** and **Prof. Fikret Sivrikaya**.

Lifemote Networks Jan – Mar 2025
Research Engineer Intern (Remote) İstanbul,Turkey

- Worked on Wi-Fi performance analysis, **network metrics**, and **automated testing tools**.

Achievements & Awards

- National University Entrance Exam (YKS): ranked 340th among 2.6 million participants **(top 0.017%)**.
- Istanbul Chamber of Industry Foundation **Merit-Based Scholarship**
- Taiwan IIPP **Summer Internship Fund**
- Erasmus+ **Summer Internship Fund**
- Bahçeşehir College High School **Full Comprehensive Scholarship**
- FIRST Robotics Competition **Chairman's Award** (2022) and **Engineering Inspiration Award** (2019)

Selected Research Projects

A Vibrotactile System for Navigation Assistance for Blind Users Jan 2026 – Present
Robotics and Mechatronics Laboratory, Prof. Çağatay Başdoğan
Target venue: IEEE World Haptics 2027.

- Developed a **navigation assistance system** integrating a stereo camera and vibrotactile vest to guide users through unfamiliar environments and warn them of obstacles.
- **Designed haptic patterns** to convey navigation information and conducted experiments with 18 participants.
- Developed **adaptive** and **predictive control strategies** for human-centered navigation.

Combined ILC & DeePC Control Framework Jul 2026 – Present
Next-generation Automated Surgical Apparatus Lab, Prof. Cheng-Wei Chen
Target venue: IEEE Conference on Decision and Control 2027.

- Developed **a hybrid control framework** integrating **Iterative Learning Control (ILC)**, **Data-Enabled Predictive Control (DeePC)**, and **Model Predictive Control (MPC)**.
- Validated the proposed control approach in **MATLAB**, incorporating input-rate constraints.

Bearcar & BeIntelli Delivery Robot Jul – Sep 2024
Distributed AI Laboratory

- Implemented and tested **SLAM algorithms with ROS 2** for the BeIntelli project's **delivery robots**.
- Developed navigation stacks for four-wheeled robots using Nav2.
- Optimized navigation parameters to improve path execution efficiency.

In-game Data Collection Pipeline Jan – Mar 2025
Lifemote Networks

- Built a **Python-based OCR pipeline** to extract and aggregate in-game Wi-Fi performance data.
- Implemented automated test setups to analyze and visualize network performance metrics.

Activities

FIRST Robotics Competition (FRC) 2018 – 2026
Mentor & Volunteer

- Won five regional awards and qualified for the **World Championship (USA)** twice.
- Organized **10+ community outreach projects** to improve **access to STEM education**.

Boğaziçi University Photography Club 2023 – 2025
***Vice President** / Board Member*

- Managed operations and organized workshops for a community of 600+ members.

BUEC – IEEE Student Branch 2022 – 2023
*Coordinator of the **Society of Women Engineers and Scientists***

Boğaziçi Airbenders UAV Team 2022 – 2023
Electronics Team Member

- Worked on **UAV electronics**, including wiring, cabling, and integration of flight control systems.

Skills & Interests

- **Programming:** Python, MATLAB, C++
- **Robotics & Simulation:** ROS 2, Nav2, Gazebo, RViz, SLAM Toolbox
- **Control & Optimization:** Simulink, MPC, ILC, DeePC, MATLAB Optimization Toolbox
- **Development:** Git, Linux
- **Languages:** Turkish (Native), English (Advanced), German (Intermediate)
- **Interests:** Mountaineering, Traveling, Analog Photography